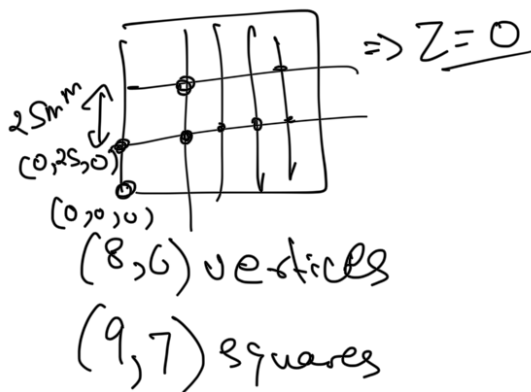


intrin matrix, dis = cv2.calibrateCamera checker
Coordinate, image

Use of matrix Algebra in CV, ML

Zhang's Method for Camera Calibration



Take
10-20 images
for checker board

Put the
Coordinates
for image and object
in the equation

Get K_{int}
and
 K_{ext}

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \frac{1}{Z} \begin{bmatrix} K_{int} & K_{ext} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$\downarrow$
edge detection

$\downarrow$
taking
a
known
object

$$K_{int} = \begin{bmatrix} \frac{f}{p_x} & 0 & c_x \\ 0 & \frac{f}{p_y} & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

$$K_{ext} = [R | t] = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix}$$

$K_{int} \Rightarrow$ is independent
of
Camera
orientation

$K_{ext} \Rightarrow$ depends
on
Camera
orientation

Used
for
mapping
image
to
real world

used to determine
orientation
of object

$f \Rightarrow$ focal
length in
mm

$p_x \Rightarrow$ distance between
two pixels in mm

$C_x, C_y \Rightarrow$ coordinates
where optical
axis cuts
the
image plane

$i \Rightarrow 1 \dots 48$

8x6 vertices $\Rightarrow$

48 points

Projection
matrix $K = K_{int} K_{ext}$

$\begin{bmatrix} u \\ v \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ z_i \\ 1 \end{bmatrix}$

$$\begin{bmatrix} C_{11} \\ C_{21} \\ C_{31} \end{bmatrix} \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} = K \begin{bmatrix} x_i \\ y_i \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ C_1 & C_2 & C_3 & C_4 \\ 1 & 1 & 1 & 1 \end{bmatrix}_{3 \times 4} \begin{bmatrix} x_i \\ y_i \\ 0 \\ 1 \end{bmatrix}_{4 \times 1}$$

$i = 1 \text{ to } 48$

$A = b$

$$\underbrace{\begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix}}_{2D \text{ } 3 \times 1} = \underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 \\ C_1 & C_2 & C_3 & C_4 \\ 1 & 1 & 1 & 1 \end{bmatrix}}_{H \text{ } 3 \times 4} \underbrace{\begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}}_{2D \text{ } 3 \times 1}$$

$$Ax = b \quad \begin{matrix} 3 \times 1 \\ 3 \times 3 \end{matrix}$$

2D to 2D $\Rightarrow$ Homography
in Homogeneous Coordinates

Create a vector with elements of H matrix

$$Ax = b$$

$$\begin{bmatrix} c_{11} \\ c_{21} \\ c_{31} \\ c_{12} \\ c_{22} \\ c_{32} \\ c_{14} \\ c_{24} \\ c_{34} \end{bmatrix} = \begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \\ h_5 \\ h_6 \\ h_7 \\ h_8 \\ h_9 \end{bmatrix}$$

$$\begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} = s_i \begin{bmatrix} h_1 & h_4 & h_7 \\ h_2 & h_5 & h_8 \\ h_3 & h_6 & h_9 \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}$$

All three equations true upto a scalar

$$\begin{aligned} h_1 x_i + h_4 y_i + h_7 &= s_i u_i \\ h_2 x_i + h_5 y_i + h_8 &= s_i v_i \\ h_3 x_i + h_6 y_i + 1 &= s_i \end{aligned}$$

Two equations

Assume $h_9 = 1$ (as h_9 is dependent parameter)

Rearranging three equations

$$h_1 x_i + h_4 y_i + h_7 - h_3 x_i - h_6 y_i = u_i$$

$$h_2 x_i + h_5 y_i + h_8 - h_3 x_i - h_6 y_i = v_i$$

Single image { Single vertex $\Rightarrow$ Two equations
 48 vertices $\Rightarrow$ 96 equations

$$\begin{bmatrix} x_1 & 0 & -x_1 & y_1 & 0 & -y_1 & 1 & 0 \\ 0 & x_1 & -x_1 & 0 & y_1 & -y_1 & 0 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & x_{48} & -x_{48} & 0 & y_{48} & -y_{48} & 0 & 1 \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \\ h_5 \\ h_6 \\ h_7 \\ h_8 \end{bmatrix} = \begin{bmatrix} u_1 \\ \cancel{u_1} \\ u_2 \\ v_2 \\ \vdots \\ u_{48} \\ v_{48} \end{bmatrix}$$

$96 \times 8 \quad 8 \times 1 \quad 96 \times 1$

$$\boxed{A h = b}$$

$96 \times 8 \quad 8 \times 1 \quad 96 \times 1$

over determined system

of equations $>$ # of Parameter

$$\begin{aligned} Ax &= b \\ x &= A^{-1}b \end{aligned}$$

Multiply by A^T both sides

$$(A^T A) h = A^T b$$

8×8

$$\begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ \vdots \\ h_8 \end{bmatrix} \leftarrow \underline{h} = \underline{(A^T A)^{-1} A^T b} \rightarrow \text{pseudo inverse solution}$$

$\underline{h}$

Zhang's Algorithm for Camera Calibration

Take
various image of
checker board

Assumes
Checker board
is
in
 $Z=0$
plane

Convert
system of
equations
to homography

Obtain
Homography
matrix

Extract
 K_{int} and
 k_{ext}
from H .

1) K_{int} is an
upper
triangular
matrix

2) Rotation
matrix
inside

k_{ext} is
orthonormal

$$k_{ext} = [R | t]$$

A is orthonormal

$$A^T A = I \text{ or } A A^T = I$$

